

Hot & Cold Work Permits

Work with ignition risk, and work without it

⚠ FOR UNDERSTANDING PURPOSES ONLY

Prasanaa Kartick T — Personal Learning Project

Flutter / Dart

Hot Work	Cold Work	Fire watch	No flame, no spark
----------	-----------	------------	--------------------

1. Hot Work Permit

Hot work is any activity that **produces or uses flame, spark, heat or an electric arc** where flammable or explosive materials might be present. Welding, cutting and grinding are the usual examples. The permit is red, tagged “Fire / Ignition”.

Activities and what makes each one different

Activity	Main hazards	Extra checklist groups
Welding / Cutting	Fire and explosion, equipment and energy, work condition, exposure and health	Spark and arc protection, enhanced fire watch, gas re-check, equipment safety, gas-cutting cylinders, fume control
Grinding / Spark Work	Abrasive disc failure, flying sparks, metal fume	Spark guard installation, grinding disc integrity
Open Flame / Heating	Open fire source, fuel gas supply, pilot light	Open-flame fire watch, heating equipment safety
Brazing / Soldering	Flux fumes, silver-solder cadmium, heat radiation	Brazing fume control, joint integrity
Hot Tapping (high risk)	Live pipe pressure, full-bore blowout, ignition at the tap	Hot-tap engineering controls, fitting weld inspection, live-line safety
Engine / Generator	Exhaust ignition, fuel spill, CO accumulation	Engine exhaust routing, fuel handling safety

General hazards and checks

- **General hazard groups:** fire and explosion (flammable gas or vapour, combustible dust, unventilated space, re-ignition from hot surfaces), equipment and energy (short circuits,

radiation, pressurised lines, live cables), work condition (confined hot space, adjacent operations, weather) and exposure and health (toxic fumes, welding smoke, eye damage, solvent vapour).

- **Nine general checklist groups:** permit and job control, work-environment safety, tools and equipment, worker competency and PPE, energy and isolation, fire and ignition control, atmosphere and exposure control, emergency readiness and simultaneous work control.
- **Fire watch.** The fire-watch role is its own item in the checklist and in the authorisation section.

2. Cold Work Permit

Cold work **produces no ignition source**: mechanical maintenance, hand-tool work, cleaning, painting, inspection and isolated electrical work. The permit is blue, tagged “Mechanical / Non-Ignition”. A strip on the form states **NO FLAME · NO SPARK · NO HEAT**, because any flame, spark or heat would turn the job into hot work.

Activities and what makes each one different

Activity	Main hazards	Extra checklist groups
Mechanical Work	Moving parts, pinch points, stored spring or hydraulic energy, pressure release	Mechanical safety, stored energy control, equipment handling
Maintenance Work	Equipment malfunction, residual process material, unexpected restart	System safety, housekeeping control
Manual Tool Work	Tool slip, struck-by, vibration, ergonomic strain	Tool safety, ergonomic control
Cleaning / Housekeeping	Slips and trips, chemical exposure, manual-handling strain	Chemical safety, slip and trip prevention
Painting / Coating	Solvent vapour, overspray, inhalation of aerosols	VOC exposure control, coating safety
Inspection / Testing	Access hazards, pressure-test failure, NDT radiation	Inspection safety, test-pressure control
Electrical Work (isolated)	Residual charge, wrong-circuit identification, tool contact	Isolation verification, live-test safety

General hazards and checks

- **Six general hazard groups:** mechanical and equipment, work environment, energy, manual handling, exposure and health, and atmosphere.
- **Eight general checklist groups:** the same as Hot Work, without the fire and ignition group.
- **Gas testing** is shown only when an atmosphere or exposure hazard is ticked.

3. Hot vs Cold at a Glance

Feature	Hot Work	Cold Work
Accent colour	Red	Blue
Ignition source	Present or possible	None allowed
General checklist groups	Nine	Eight
Fire watch	Required item	Not applicable
Gas testing	Shown when a fire or gas hazard is ticked	Shown when an atmosphere or exposure hazard is ticked
Section 1 and Section 5	Standard	Identical to Hot Work, in blue